

ANALYSTLAB AFRICA

Experience Lab Internship Programme



HealthConnect Clinic Data Science Track Project Plan

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Project: HealthConnect Clinic Experience Lab – Improving Patient Appointment Attendance and
Healthcare Support Using Data and AI

Track: Data Science

Prepared for the AnalystLab Africa Experience Lab
Week 4 Submission – Supporting Document

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1. Introduction

This Project Plan sets out how the Data Science track intends to progress the HealthConnect no-show prediction problem from the initial definition completed in Week 4 through to a final, evaluated model and presentation. It is a supporting planning document for this track and complements, rather than replaces, the Project Management track's own Project Initiation Package for the shared HealthConnect project.

The plan follows the five-stage project journey set out in the Week 4 brief: Problem Understanding, Analysis and Solution Design, Development, Testing and Refinement, and Final Presentation. Structuring the plan around a work breakdown structure and a phased timeline follows standard project planning practice (Project Management Institute [PMI], 2021).

2. Project Objectives

The Data Science track's objective is to produce a validated, well-documented no-show prediction model, together with the evidence and reasoning behind every major modelling decision, so that the model can be responsibly integrated into HealthConnect Clinic's patient-support workflow by the Machine Learning Engineering and Project Management tracks.

3. Scope Statement

- **In scope:** defining and validating a no-show prediction problem using the HealthConnect Appointment Dataset; data-quality assessment; feature engineering and selection; training, tuning and evaluating classification models; and documenting all decisions for cross-track use.
- **Out of scope:** building or deploying a production system (the responsibility of the Machine Learning Engineering track); designing the Generative AI assistant; and managing the shared HealthConnect project timeline or stakeholder communications (the responsibility of the Project Management track).

4. Work Breakdown Structure

WBS ID	Work Package	Stage
1.0	Problem Understanding	Week 4
1.1	Review project resources and business scenario	Week 4
1.2	Define the machine learning problem and target variable	Week 4
1.3	Identify candidate input features and initial evidence	Week 4
1.4	Prepare Machine Learning Problem Definition Document and supporting registers	Week 4
2.0	Analysis and Solution Design	Week 5
2.1	Resolve open data-quality questions (e.g., waiting_time_minutes)	Week 5
2.2	Conduct formal statistical and correlation analysis	Week 5
2.3	Complete feature encoding and scaling	Week 5
2.4	Build and evaluate a baseline logistic regression model	Week 5

3.0	Development	Week 6
3.1	Train candidate ensemble models (e.g., random forest, gradient boosting)	Week 6
3.2	Tune hyperparameters using the validation set	Week 6
3.3	Compare models against agreed evaluation metrics	Week 6
3.4	Conduct a subgroup fairness review	Week 6
4.0	Testing and Refinement	Week 7
4.1	Evaluate the selected model on the held-out test set	Week 7
4.2	Conduct error analysis and refine features or model choice	Week 7
4.3	Document the final model and its limitations	Week 7
5.0	Final Presentation	Week 8
5.1	Prepare final report and presentation materials	Week 8
5.2	Coordinate a cross-track integration walkthrough	Week 8
5.3	Deliver final presentation and update GitHub/portfolio	Week 8

5. High-Level Timeline

Week	Phase	Key Activities	Deliverable
Week 4	Problem Understanding	Resource review; problem, target and feature definition	ML Problem Definition Document, Week 4 Project Summary, registers
Week 5	Analysis and Solution Design	Data-quality resolution; statistical analysis; baseline model	Baseline model and analysis notebook
Week 6	Development	Candidate model training, tuning and fairness review	Candidate model comparison report
Week 7	Testing and Refinement	Test-set evaluation, error analysis, refinement	Final validated model and documentation
Week 8	Final Presentation	Reporting, cross-track integration, presentation	Final presentation and portfolio update

6. Milestones

- **M1 (Week 4):** Machine Learning Problem Definition Document and supporting registers completed.
- **M2 (Week 5):** Feature set finalised and baseline model trained.
- **M3 (Week 6):** Candidate model selected following comparison and fairness review.
- **M4 (Week 7):** Final model validated on the test set and fully documented.
- **M5 (Week 8):** Final presentation delivered and portfolio updated.

7. Roles, Responsibilities and Cross-Track Coordination

Within this plan, the Data Science track is responsible for all data analysis, feature engineering and model development activity described above. Coordination with the other HealthConnect tracks is required at several points:

- **Project Management:** provides the overall project timeline, stakeholder register and communication approach that this track's milestones must align with.
- **Machine Learning Engineering:** depends on this track's target variable, feature set and modelling approach to design the production system architecture (see the accompanying ML System Architecture Design and ML Workflow Design documents).
- **Data Analytics:** may draw on the same dataset and business questions; findings should be cross-checked for consistency where they overlap.
- **Generative AI:** shares the Clinic Knowledge Base used in this track to interpret appointment outcomes, and may reference this track's findings when describing appointment-related risk to users.

References

- AnalystLab Africa. (2026). *AnalystLab Africa Experience Lab – Week 4: HealthConnect project kickoff & problem understanding*. Assignment brief.
- AnalystLab Africa. (2026). *HealthConnect Clinic: Machine learning problem definition document*. Week 4 submission.
- Project Management Institute. (2021). *A guide to the project management body of knowledge (PMBOK guide)* (7th ed.). Project Management Institute.